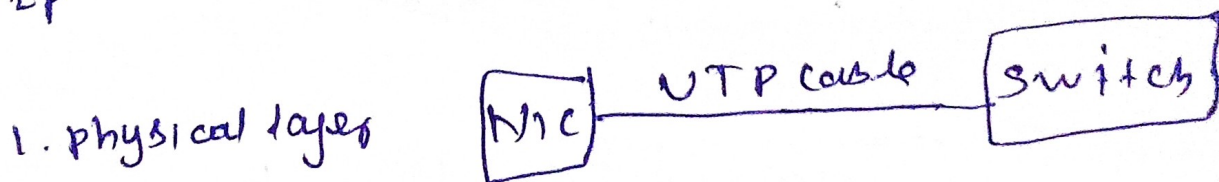
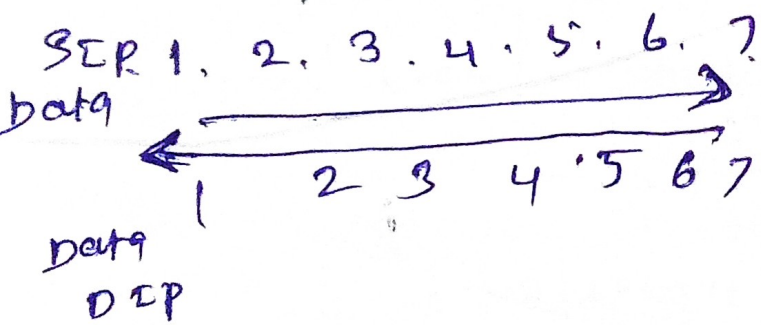


OSI

1. Open system interconnection
2. OSI standard methods
3. All network device controlling (OSI)
4. 7 layers available.

1. Application Layer, 2. presentation layer, 3. Session
4. Transport 5. Network layer, 6. Data Link layer
7. Physical layer.



1. How ping works OSI Layer.
2. ping (packet internet groper)
3. if we ping.

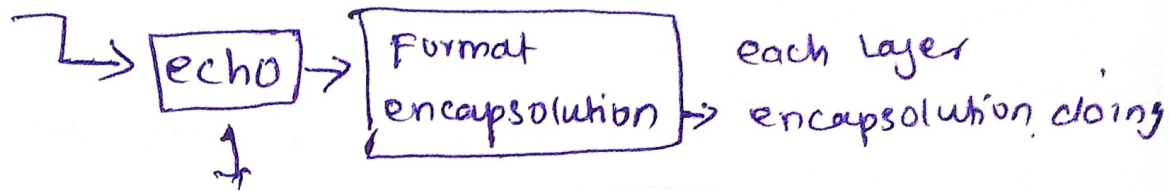
generating application layer it's called **echo**
size, 32 byte, icmp request echo data
generate again.

Application Layer

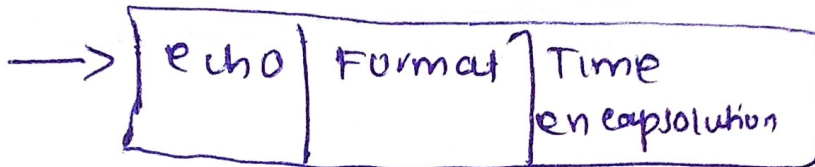
Application: ping → Generating echo (data) with help of

↓
Icmp (internet control protocol)

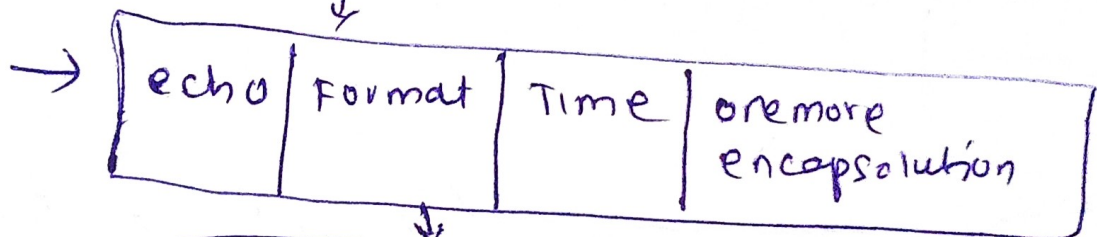
Presentation



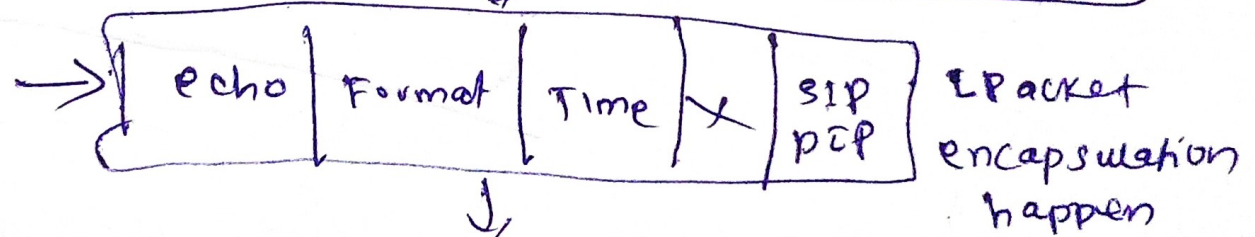
Session



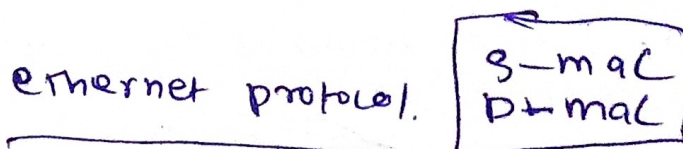
Transport



Network



Data Link Layer



↳ does not know D-mac.

2 not happen encapsulation

3. Solution

ARP : ADDRESS Resolution protocol.

↳ it is a protocol. it use to find D-mac with help of destination - IP

Ethernet called

↓ Request to find D-mac.

ARP →

New Generated

ARP TABLE Req

E am: 50.0.0.10
mac: 00A
who: 50.0.0.20
Give a mac

S. mac } encapsulation
D. mac ? }

D. mac. Does not know

So, FFFFFFFF Data Generating to Broadcast Communication.

↓ mac

↓ ARP Request Broadcast

Physical

echo	request	time	x	D-IP	S-mg
				S-IP	D-mg

ARP Request Broadcast

ARP Reply

E am: 50.20
mac: 00B
who: 00B

Noted

ARP Table

① mac: 00B

② Send to ethernet port)

③ ethernet port- mac Add Frame encapsulation)

00B Reply

↓ unicast